

Lecture 19: one more unconstrained optimization e.g.
18/4/2 intro to optimization under 1 constraint

Consider the f^n $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, defined by

$$f(x, y) = x^2 + \frac{x}{2} + x \cos(2y)$$

- a) Find all the critical points of the f^n f
b) Determine the nature (i.e. point of local max/min, saddle point) of the critical points $(0, \frac{\pi}{3})$ and $(\frac{1}{4}, \frac{\pi}{2})$
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we seek (x, y) s.t. $\nabla f(x, y) = 0$

$$\begin{cases} f_x = 0 \\ f_y = 0 \end{cases} \Rightarrow \begin{cases} 2x + \frac{1}{2} + \cos(2y) = 0 \\ -2x \sin(2y) = 0 \end{cases}$$

$$\begin{cases} 2x + \frac{1}{2} + \cos(2y) = 0 & \textcircled{1} \\ -2x \sin(2y) = 0 & \textcircled{2} \end{cases}$$

$$\textcircled{2} \Rightarrow x=0 \text{ or } \sin(2y)=0$$

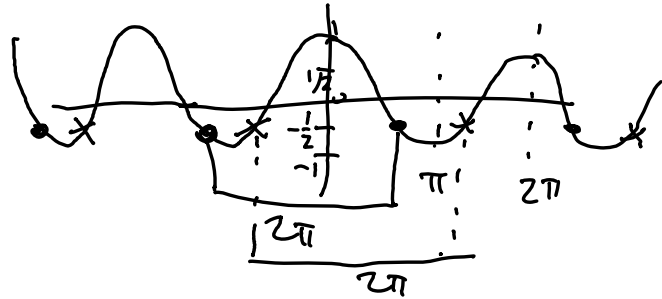
$$\text{If } \boxed{x=0}, \textcircled{1} \Rightarrow 2 \cdot 0 + \frac{1}{2} + \cos(2y) = 0 \Rightarrow \cos(2y) = -\frac{1}{2}$$

$$\arccos\left(-\frac{1}{2}\right) = \frac{2}{3}\pi$$

$$\left. \begin{aligned} 2y &= \frac{2}{3}\pi + k \cdot 2\pi \quad k \in \mathbb{Z} \\ 2y &= -\frac{2}{3}\pi + k \cdot 2\pi \quad k \in \mathbb{Z} \end{aligned} \right\}$$

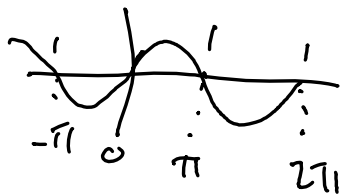
$$\Rightarrow y = \frac{\pi}{3} + k\pi \quad \text{or} \quad -\frac{\pi}{3} + k\pi$$

$$\text{CP: } \left(0, \frac{\pi}{3} + k\pi\right), \left(0, -\frac{\pi}{3} + k\pi\right) \quad k \in \mathbb{Z}$$



$$\text{if } \sin(2y) = 0$$

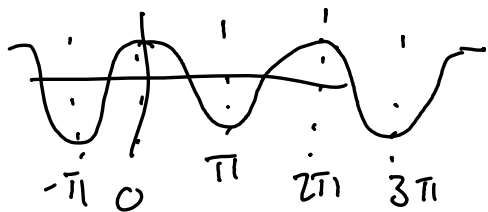
$$\begin{cases} 2x + \frac{1}{2} + \cos(2y) = 0 & \textcircled{1} \\ -2x \sin(2y) = 0 & \textcircled{2} \end{cases}$$



$$2y = n\pi \quad n \in \mathbb{Z}$$

$$y = \frac{n}{2}\pi$$

What happens in $\textcircled{1}$ now? consider $\cos(n\pi)$?



if n is even $[n = 2m \quad m \in \mathbb{Z}]$

$$\cos(n\pi) = 1$$

if n is odd $[n = 2m+1 \quad m \in \mathbb{Z}]$

$$\cos(n\pi) = -1$$

When $n = 2m$, $\cos(n\pi) = \cos(2m\pi) = 1$. $\textcircled{1} \Rightarrow 2x + \frac{1}{2} + 1 = 0$
 $\Rightarrow x = (-1 - \frac{1}{2})/2 = -\frac{3}{2}/2 = -\frac{3}{4}$. cp: $(-\frac{3}{4}, \frac{2m}{2}\pi) = (-\frac{3}{4}, m\pi)$

When $n = 2m+1$, $\cos((2m+1)\pi) = -1$. $\textcircled{1} \Rightarrow 2x + \frac{1}{2} - 1 = 0$
 $\Rightarrow x = (-\frac{1}{2})/2 = -\frac{1}{2}/2 = -\frac{1}{4}$. cp: $(-\frac{1}{4}, \frac{2m+1}{2}\pi) = (-\frac{1}{4}, m\pi + \frac{\pi}{2})$

$$\text{cps } \left(0, \frac{\pi}{3} + k\pi\right), \left(0, -\frac{\pi}{3} + k\pi\right), \left(-\frac{3}{4}, m\pi\right), \left(\frac{1}{4}, m\pi + \frac{\pi}{2}\right)$$

$$k, m \in \mathbb{Z}$$

b) nature of $\left(0, \frac{\pi}{3}\right)$ and $\left(\frac{1}{4}, \frac{\pi}{2}\right)$

$$f_{xx} = 2 \quad f_{xy} = -2\sin(2y)$$

$$= f_{yx}$$

(since f is of class C^2)

$$f_x = 2x + \frac{1}{2} + \cos(2y)$$

$$f_y = -2x\sin(2y)$$

$$f_{yy} = -2x\cos(2y) \cdot 2$$

$$= -4x\cos(2y)$$

$$D\left(0, \frac{\pi}{3}\right) = \begin{vmatrix} f_{xx}(0, \pi/3) & f_{xy}(0, \pi/3) \\ f_{xy}(0, \pi/3) & f_{yy}(0, \pi/3) \end{vmatrix} = \begin{vmatrix} 2 & -2\sin(2\pi/3) \\ -2\sin(2\pi/3) & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & -2 \cdot \frac{\sqrt{3}}{2} \\ -2 \cdot \frac{\sqrt{3}}{2} & 0 \end{vmatrix} = \begin{vmatrix} 2 & -\sqrt{3} \\ -\sqrt{3} & 0 \end{vmatrix} = 2 \cdot 0 - (-\sqrt{3} \cdot -\sqrt{3})$$

$$= 0 - 3 = -3 < 0$$

thus $\left(0, \frac{\pi}{3}\right)$ is a saddle point

$$\left(\frac{1}{4}, \frac{\pi}{2}\right) \quad f_{xx} = 2 \quad f_{xy} = -2 \sin(2y) \quad f_{yy} = -4x \cos(2y)$$

$$D\left(\frac{1}{4}, \frac{\pi}{2}\right) = \begin{vmatrix} f_{xx}\left(\frac{1}{4}, \frac{\pi}{2}\right) & f_{xy}\left(\frac{1}{4}, \frac{\pi}{2}\right) \\ f_{xy}\left(\frac{1}{4}, \frac{\pi}{2}\right) & f_{yy}\left(\frac{1}{4}, \frac{\pi}{2}\right) \end{vmatrix} = \begin{vmatrix} 2 & -2 \sin\left(2 \cdot \frac{\pi}{2}\right) \\ -2 \sin\left(2 \cdot \frac{\pi}{2}\right) & -4 \cdot \frac{1}{4} \cdot \cos\left(2 \cdot \frac{\pi}{2}\right) \end{vmatrix}$$

$$= \begin{vmatrix} 2 & -2 \cdot 0 \\ -2 \cdot 0 & -1 \cdot (-1) \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = 2 \cdot 1 - 0 \cdot 0 = 2 > 0$$

$$f_{xx} = 2 > 0 \quad (\text{or } f_{yy} = 1 > 0)$$

$\left(\frac{1}{4}, \frac{\pi}{2}\right)$ is a local minimum

Optimizing under one equality constraint $\begin{cases} \text{max-min} \\ \text{II} \end{cases}$

we'll deal with problems of the following form:

- Find the max/min of $f(x,y)$ subject to $g(x,y)=0$
In other words, we're looking for the values of (x,y) s.t. $g(x,y)=0$ and the output of $f(x,y)$ is maximal/minimal
- Find the max/min of $f(x,y,z)$ subject to $h(x,y,z)=0$

What does this mean visually? let's work with $f(x,y)$ subject to $g(x,y)=0$

$$f(x,y) = x^2 - y^2$$

$$g(x,y) = x^2 + y^2 - 1 = 0$$

look at geogebra...

Theorem: Let $f, g: \mathbb{R}^2 \rightarrow \mathbb{R}$ be functions of class C^1 (i.e. first derivative is continuous) and let (a, b) be such that $g(a, b) = 0$ and $\nabla g(a, b) \neq (0, 0)$.

If (a, b) is a solution to the problem "find max/min of $f(x, y)$ subject to $g(x, y) = 0$ ", then

$$\nabla f(a, b) \parallel \nabla g(a, b) \Leftrightarrow \exists \lambda \in \mathbb{R} \text{ s.t. } \nabla f(a, b) = \lambda \nabla g(a, b)$$

(grad of f @ (a, b) is
parallel to g @ (a, b))

$$\Leftrightarrow \exists \lambda \in \mathbb{R} \text{ s.t. } \begin{cases} f_x(a, b) = \lambda g_x(a, b) \\ f_y(a, b) = \lambda g_y(a, b) \end{cases}$$

We'll discuss the proof later

!

How will we use this? "find max/min at $f(x,y)$ subject to $g(x,y)=0$ "

Step 1: solve for (x,y) and λ s.t.
$$\begin{cases} f_x(x,y) = \lambda g_x(x,y) \\ f_y(x,y) = \lambda g_y(x,y) \\ g(x,y) = 0 \end{cases}$$

Step 2: consider all the (x,y) found in step 1 and evaluate the output of $f(x,y)$ for each
e.g. $f(1,1) = 5$, $f(3,4) = 2$, $f(7,9) = -2$, $f(10,2) = 5$
biggest neither smallest bigges

Step 3: decide, based on outputs obtained in step 2, which points (x,y) lead to maximal and minimal values of f .

e.g. $(1,1)$ and $(10,2)$ are points of maximum
 $(7,9)$ is a point of minimum

• Find the max/min of $f(x,y,z)$ subject to $h(x,y,z)=0$

Theorem: Let $f, h: \mathbb{R}^3 \rightarrow \mathbb{R}$ be f^{ns} of class C^1 and let (a,b,c) be s.t. $h(a,b,c)=0$ and $\nabla h(a,b,c) \neq (0,0,0)$. If (a,b,c) is a solution of the above problem, then

$$\nabla f(a,b,c) \parallel \nabla h(a,b,c) \Leftrightarrow \exists \lambda \in \mathbb{R} \text{ s.t. } \nabla f(a,b,c) = \lambda \nabla h(a,b,c)$$

$$\Leftrightarrow \exists \lambda \in \mathbb{R} \text{ s.t. } \begin{cases} f_x(a,b,c) = \lambda h_x(a,b,c) \\ f_y(a,b,c) = \lambda h_y(a,b,c) \\ f_z(a,b,c) = \lambda h_z(a,b,c) \end{cases}$$

How will we use this? "find max/min of $f(x,y,z)$ subject to $h(x,y,z)=0$ "

Step 1: solve for (x,y,z) and λ s.t.
$$\begin{cases} f_x(x,y,z) = \lambda h_x(x,y,z) \\ f_y(x,y,z) = \lambda h_y(x,y,z) \\ f_z(x,y,z) = \lambda h_z(x,y,z) \\ h(x,y,z) = 0 \end{cases}$$

Step 2: consider all the (x,y,z) found in step 1
and evaluate the output of $f(x,y,z)$ for each
e.g. $f(1,2,3)=10$, $f(4,5,6)=-15$, $f(7,8,9)=-15$

Step 3: decide, based on ^{biggest} ^{smallest} outputs obtained in step 2,
which points (x,y,z) lead to maximal and
minimal values of f .

e.g. $(1,2,3)$ is a point of maximum
 $(4,5,6)$ and $(7,8,9)$ are points of minimum

this method is called the
"method of Lagrange multipliers"

λ - Lagrange multiplier

max-min III

2 equality constraints

$$g(x,y)=0 \quad k(x,y)=0$$

$$\nabla f(x,y) = \lambda \nabla g(x,y) + \mu \nabla k(x,y)$$

Lagrange multipliers